

ID2221 - Data-Intensive Computing (Dec. 19, 2024)

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No aids are allowed

The exam consists of 20 multiple-choice questions, each with only one correct answer, and each question has one point. The final grading for the exam is as follows:

A: $x \geq 18$, **B:** $18 < x \leq 16$, **C:** $16 < x \leq 14$, **D:** $14 < x \leq 12$, **E:** $12 < x \leq 10$, **F:** $x < 10$

1. How does GFS distinguish between up-to-date and stale replicas?

- (a) By periodically comparing data across all chunkservers. **granted.**
 - (b) **By assigning version numbers to chunks and updating them when leases are**
 - (c) By storing replicas in different racks for cross-verification.
 - (d) By maintaining logs of all write operations.
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2. What is the purpose of the Chubby lock service in BigTable?

- (a) To replicate data across tablet servers.
 - (b) To manage read and write operations.
 - (c) **To store BigTable schema information and access control lists.**
 - (d) To cache tablet locations.
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3. What is a primary benefit of the column-oriented data model compared to the row-oriented model?

- (a) Better integration with SQL queries.
 - (b) Improved storage of hierarchical data.
 - (c) **Faster operations on specific columns of data.**
 - (d) Simplified schema management.
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4. When a new node is added to a Cassandra cluster that uses consistent hashing, what happens to the existing data?

- (a) All existing data is rehashed and redistributed across all nodes, including the new node.
- (b) The new node takes over responsibility for a portion of the data, but no data is moved from existing nodes.
- (c) **Only the data that falls within the new node's hash range is moved from existing nodes to the new node.**
- (d) The new node remains idle until a sufficient amount of new data is inserted to warrant its use.

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5. What trade-off does BigTable make by prioritizing strong consistency over availability?
- (a) **Temporary unavailability of data when a tablet server fails.**
 - (b) Increased redundancy.
 - (c) Reduced read/write latency.
 - (d) Simplified replication process.
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6. What is the role of the shuffle and sort phase in MapReduce?
- (a) To execute the map tasks in parallel.
 - (b) To divide the input data into smaller splits.
 - (c) To replicate the reduce tasks across multiple nodes.
 - (d) **To group and consolidate intermediate key-value pairs by keys.**
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7. In a reduce-side join in MapReduce, what must be ensured before performing the join operation?
- (a) **Data is sorted and grouped by the join key.**
 - (b) Both datasets are partitioned into equal-sized splits.
 - (c) One of the datasets is small enough to fit in memory.
 - (d) Intermediate data is written to the master node.
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8. Which of the following best describes the role of the lineage graph in Spark?
- (a) **It represents the sequence of RDD operations for fault tolerance.**
 - (b) It is used to schedule tasks across executors.
 - (c) It determines the partitioning of data across the cluster.
 - (d) It handles the caching of RDDs in memory.
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9. What is the primary advantage of caching an RDD in Spark?
- (a) It ensures the RDD is distributed across all executors.
 - (b) It allows the RDD to be checkpointed to disk.
 - (c) It reduces the number of partitions in the RDD.
 - (d) **It avoids re-evaluating the RDD for subsequent actions.**
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10. What is the output of the `groupBy` operation in a Spark `DataFrame`?
- (a) **A new `DataFrame` with grouped data ready for aggregation.**
 - (b) A `Dataset` containing only distinct rows.
 - (c) A `DataFrame` with columns renamed based on the grouping keys.
 - (d) A new RDD with keys representing grouping attributes.

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11. What is the key advantage of using Datasets over DataFrames?
- (a) Datasets allow untyped transformations.
 - (b) Datasets automatically cache all transformations.
 - (c) **Datasets provide compile-time type safety.**
 - (d) All of the above.
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12. What is the purpose of watermarks in stream processing?
- (a) To synchronize event time and processing time.
 - (b) To buffer data for micro-batch processing.
 - (c) To identify partitions within a topic.
 - (d) **To provide a threshold indicating how long the system waits for late events.**
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13. Which of the following best describes a sliding window in stream processing?
- (a) A window that retains a fixed number of tuples before discarding old ones.
 - (b) A window that supports batch operations by evicting all tuples after processing.
 - (c) **A window that allows overlapping buffers for incremental processing.**
 - (d) A window that aggregates data over a fixed period.
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14. In Kafka, what is the significance of the offset associated with each message?
- (a) **It uniquely identifies the position of a message within a partition.**
 - (b) It ensures at-least-once delivery of messages.
 - (c) It guarantees the total ordering of messages across partitions.
 - (d) It enables the replication of messages across brokers.
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15. Why is checkpointing required for stateful operations in Spark Streaming?
- (a) To persist data to external storage for later analysis.
 - (b) To store RDD transformations for fault tolerance.
 - (c) **To save intermediate states and ensure recovery in case of failures.**
 - (d) To enable window-based transformations.
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16. What is the main difference between Spark Streaming and Structured Streaming?
- (a) **Spark Streaming uses DStreams, while Structured Streaming uses continuous tables.**
 - (b) Spark Streaming supports window operations, while Structured Streaming does not.
 - (c) Spark Streaming processes data in real-time, while Structured Streaming processes in batches.
 - (d) Spark Streaming handles late data natively, while Structured Streaming does not.

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17. What is the main challenge with using edge-cut partitioning for natural graphs?
- (a) It does not balance the number of vertices across partitions.
 - (b) **It performs poorly on skewed power-law degree distributions.**
 - (c) It requires duplicating all edges across partitions.
 - (d) It is incompatible with iterative graph algorithms.
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18. Why is asynchronous computation in GraphLab often faster than synchronous models like Pregel?
- (a) It processes vertices sequentially rather than in parallel.
 - (b) It requires fewer iterations to converge on large graphs.
 - (c) It avoids partitioning the graph altogether
 - (d) **It eliminates the need for global synchronization between supersteps.**
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19. What is a key limitation of max-min fairness in resource allocation?
- (a) It does not guarantee optimal resource utilization.
 - (b) **It cannot handle scenarios with heterogeneous resources.**
 - (c) It prioritizes certain users over others based on task requirements.
 - (d) It leads to frequent re-allocation of resources.
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20. How does Delta Lake ensure data reliability in its tables?
- (a) By replicating data across multiple storage layers.
 - (b) By caching all operations in memory.
 - (c) By restricting updates to batch processes only.
 - (d) **Through a DeltaLog that tracks changes and ensures consistency.**